

Jacob Salminen

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LinkedIn: www.linkedin.com/in/jacob-salminen-124a50129/

GitHub: <https://github.com/JacobSal>

EDUCATION

University of Florida Herbert Wertheim College of Engineering

Gainesville, FL

Bachelor of Science in Chemical Engineering

August 2015-December 2019

Minor in Mathematics

Magna Cum Laude

GPA: 3.64

University of Florida: Department of Biomedical Engineering

Gainesville, FL

Doctor of Philosophy in Biomedical Engineering

August 2020-Present

Focus: Neural Engineering and Biomechanics

Advisor: Dr. Daniel Ferris (Human Neuromechanics Laboratory)

GPA: 3.85

Course work encompasses topics in neural engineering, software engineering, computational biomechanics, artificial intelligence, and statistics.

RESEARCH & EXPERIENCE

UF Biomedical Engineering

August 2020-Present

Human Neuromechanics Laboratory – Graduate Assistant (PI: Daniel Ferris Ph.D.)

- Graduate Assistantship under the mentorship of Daniel Ferris Ph.D. in the Human Neuromechanics Laboratory. Research foci include EEG, neural engineering, supercomputing (HiPerGator), signal processing, and biomechanics. I hope my studies can help increase our knowledge of physical therapy, mental health, rehabilitation, neurophysiology, artificial intelligence.
- **Proposed Dissertation Work** (Mind in Motion, NIA 1U01AG061389-01): Define effects of age-related decline using biomechanical, EEG, MRI, fNIRS, and clinical outcome measures. I work with several independent researchers to collect high-density EEG data of young and older adults (N=165) while they walk on different treadmill terrain surfaces. We use inertial measurement units and force sensors inserted into the sole of the shoe to quantify changes in locomotion across the different surfaces. We use the high-density EEG measures to understand the correlation of electrocortical dynamics to mobility ability.
 - **Project 1:** Develop MATLAB scripts to process the large and diverse dataset (N=165) available through the study. I designed pipelines to clean the EEG data using signal-processing techniques. We extract source level (within the brain) signals from the EEG using the AMICA independent component analysis algorithm and custom head models created from subject MRIs and 3D scanned electrode locations. Using the source localized signals I then create descriptive graphs and reports. I have scripts to create automatically generated reports for all subjects.
 - **Project 2:** Develop MATLAB scripts to analyze the effective connectivity between source localized EEG signals within the brain. Effective connectivity measures provide insight into the relative causation of one brain area on a different brain area. I use this understanding to test hypotheses on the control system underlying mobility ability.
 - **Project 3:** EEG generates large amounts of data, which can slow the timeline to producing publishable results. A goal of mine was to improve the speed and memory efficiency by which EEG data processes. I developed a pipeline that can improve the processing time for EEG data using the UF HiPerGator. I used bash, slurm, and MATLAB to decrease the calculation time for our current dataset from ~30days to ~1day. After publication, the GitHub repository containing the code will be available.
 - **Collaborative Connectivity Projects:** I use my connectivity pipelines to test hypotheses related to electrocortical dynamics during table tennis play (lead: Amanda Studnicki) and split belt treadmill walking (lead: Noelle Jacobsen). I work with lab members that use mobile brain body imaging with EEG to develop statistics and graphs that represent the effective connectivity patterns and trends during table tennis play and split belt walking.

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BestBuy Health & Technology Center

June 2021-January 2022

Applied Machine Learning - Graduate Intern

- Work on an agile, multidisciplinary team with full-time employees in engineering, data science, and quality. Exposure into the development of medical and wellness devices, as well as to Best Buy's emerging health business.
- **Project Goal:** Determine if a chest or wrist IMU and a clinically informed algorithm can detect falls in older adults. Specifically determine a biomechanical marker for falls using a barometer sensor.
 - Use MATLAB to train a decision tree using accelerometer and barometer biomarkers for falls.
 - Collect fall data using a wrist and chest mounted IMU.
- **Project Outcome:** By including a barometer sensor into the established fall sensing algorithm, I improved the machine learning algorithm's ability to detect falls by

UF Respiratory Neurobiology and Physiology

OPS Laboratory Technician

January 2020-August 2020

- Developed machine learning pipelines (SVM, Decision Tree, KNN, & CNN) using Python (scikit-learn, PyTorch, opencv, & numba) to detect immunofluorescent serotonin in images of spinal cord sections.

Physiology laboratory/ Research Assistant

October 2017-January 2020

- **Undergraduate Thesis:** Designed an algorithm in Python. The Algorithm includes customized Fourier transformations and convolutional techniques to detect and quantify different cell types in images of spinal cords.

University of Florida & City of Gainesville

June 2017-July 2018 & May 2019-December 2020

Lifeguard

LEADERSHIP & INVOLVEMENT

University of Florida, Biomedical Engineering Department

August 2020-Present

Girls with Nerve, Member

- Present and participate in activities to teach underprivileged children about neuroscience, neural engineering, and research. (e.g., presentations, "bionic hands" made of straws and strings).

University of Florida, Biomedical Engineering Department

August 2020-Present

Graduate Student Counsel, Spring Recruitment Chair & Member

- Coordinate events and activities to recruit potential Graduate Students (e.g., organizing events, staff, and travel).
- Monthly meetings to discuss actions to improve the direction and sustainability of the department and its faculty/Students

SKILLS & EXTRACURRICULAR

- | | |
|--------------------------------|--------------------------------------|
| • Python (Expert) | • Slurm, Bash, Linux/UNIX, Intel MPI |
| • MATLAB (Expert) | (Proficient) |
| • C/C#/C++ & CUDA (Proficient) | • Guitar, Games, and Mountain Biking |

PUBLICATIONS, POSTERS & ABSTRACTS

Streeter KA, Wollman LB, Fusco FA, Poirier TP, Salminen JS, Fuller DD, "Electrical activation of phrenic afferent neurons evokes phrenic motor plasticity following cervical spinal cord injury", (2019) Experimental biology Conference Abstract; Department of Physical Therapy, University of Florida.

Salminen JS, Sunshine M, Cowen G, Otto KJ, Fusco AF, Fuller DD, "Automated Quantification of Microglia Using a Novel Algorithm and CNN-UNet in a Spinal Cord Injured Mouse Model", (2019) University of Florida Honors Thesis; Department of Physical Therapy, University of Florida. <https://ufdc.ufl.edu/AA00069737/00001>

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Salminen JS, Seven YB, Luca E, Gordon MS, “*Assessing Rat Spinal Cord Serotonin Distribution Using Machine Learning*”, (2021) Manuscript; Department of Physical Therapy, University of Florida.

Liu C, Downey RJ, Salminen JS, Ferris DP, “*Neural oscillations during uneven terrain walking*”, (2023) IEEE Neural Engineering, Conference Abstract, University of Florida.

Liu C, Downey RJ, Salminen JS, Rojas SA, Richer N, Pliner EM, Hwang J, Cruz-Almeida Y, Manini TM, Hass CJ, Seidler RD, Clark DJ, Ferris DP. Electrical Brain Activity during Human Walking with Parametric Variations in Terrain Unevenness and Walking Speed. bioRxiv [Preprint]. 2023 Aug 2:2023.07.31.551289. doi: [10.1101/2023.07.31.551289](https://doi.org/10.1101/2023.07.31.551289). PMID: 37577540; PMCID: PMC10418077.

WORKSHOPS

Center for Respiratory Research and Rehabilitation Workshop, Hosted by University of Florida in Gainesville, FL (2019)

Neurotherapeutic Intermittent Hypoxia Workshop, Hosted by University of Florida in Jacksonville, FL (2020)

Mobile Brain Body Imaging Workshop & Conference, Hosted by Swartz Center for Computational Neuroscience in San Diego, CA (2022)

HONORS & AWARDS

Graduate School Preeminence Award, University of Florida (2020)